

Practice Paper 1: Multiple Choice Questions (Previous Year Pattern)

Covers: Probability, Dist, Matrices, Limits, Co-ord Geo, Calculus, Algebra, AP/GP.

1. If A is a square matrix such that $A^2 = I$, then $(I-A)^3 + (I+A)^3$ is equal to: a) $2I$, b) $4I$, c) $6I$, d) $8I$

2. The value of $\lim_{x \rightarrow 0} (\sin x / x)$ is:

a) 0, b) 1, c) ∞ , d) Undefined

3. The sum of the first 10 terms of an AP with $a=2$ and $d=3$ is:

a) 150, b) 155, c) 160, d) 165

4. For a fair die, the probability of getting an even number is:

a) $1/6$, b) $1/3$, c) $1/2$, d) 1

5. The eccentricity of a circle is:

a) 0, b) 1, c) <1 , d) >1

6. Finding Asymptotes using Limits

Let $f(x) = \frac{2x^2 - 5x + 2}{x^2 - 4}$. Which of the following statements is true?

- A) $f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 0$.
- B) $f(x)$ has a vertical asymptote at $x = -2$ and a horizontal asymptote at $y = 2$.
- C) $f(x)$ has vertical asymptotes at $x = 2$ and $x = -2$, and a horizontal asymptote at $y = 2$.
- D) $f(x)$ has a vertical asymptote at $x = -2$ and a horizontal asymptote at $y = 0$.

7. Limit Involving Absolute Value

Evaluate the limit: $\lim_{x \rightarrow 3^-} \frac{|x - 3|}{x^2 - 9}$

- A) $-\frac{1}{6}$
- B) $\frac{1}{6}$
- C) 0
- D) The limit does not exist.

